

(TR-104) PYTHON WITH AI/ML

Training Day 54 Report

Date: 26 March 2026

Topic Covered: Workflow Automation, MCP Integration & Smart Task Agent Development

The fifty-fourth day of training focused on studying workflow automation models, integrating MCP with Ollama for AI agent workflows, and exploring automation concepts using n8n.

- Studied different workflow models including sequential, parallel, conditional, iterative, and event-driven workflows.
- Explored the Praison AI framework for implementing workflows using MCP and Ollama integration.
- Analyzed the architecture where Ollama acts as an AI agent interacting with tools through FastMCP.
- Built an MCP server integrated with multiple tools for content generation, memory handling, and task history management.
- Integrated Ollama (llama3.2) with MCP for tool calling and AI-driven task execution.
- Implemented generate_content tool for automated social media content generation.
- Designed structured prompts for generating captions, hashtags, and scheduling-related outputs.
- Implemented memory storage using JSON files for persistent session management.
- Created tools for saving, retrieving, listing, deleting, and clearing memory records.
- Developed task history tracking system with timestamp-based logging.
- Maintained separation between conversational memory and task history modules for better architecture.
- Explored workflow automation concepts using n8n.
- Configured local n8n setup and connected Ollama through HTTP endpoints.

- Built workflows using trigger nodes and “Message a Model” nodes for AI interactions.
- Tested automated workflows using different prompts for content generation and task planning.
- Strengthened understanding of workflow orchestration and AI agent automation pipelines.
- Improved understanding of nodes, triggers, actions, utilities, and AI components inside n8n workflows.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/agents>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/building-a-smart-task-agent-using-mcp-and-ollama-8ff3ea71ea3d>